Supervyse OPF Redfish VirtualMedia User Guide

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Revision History

Revision Number	Description	Author	Reviewer	Release Date
1.0	Initial release.	George Huang	Victor Wu	Feb, 2025

1 Introduction

1.1 Overview

This document is intended to guide developers, users, and testers understand how to effectively use the Virtual Media service in Redfish for Supervyse OPF.

1.2 Related Information

This document references the following publications and sources of information, which may also be useful to you:

- *Supervyse OPF Redfish API User Guide, 1.4, Insyde Software, 2024*
- *Supervyse OPF Redfish CertificateService User Guide, 1.0, Insyde Software, 2025*

1.3 Terms

The following terms are used throughout this document:

cURL

A command-line tool for transferring data using various network protocols.

2 Virtual Media Overview

2.1 What is Virtual Media

The Virtual Media service allows users to **remotely mount digital media**—such as ISO files, operating system images, firmware updates, or other software—**through the BMC to the host system**.

2.2 Virtual Media Collection

Users can read a list of virtual media devices available on the system by sending a **GET** request to: `/redfish/v1/Managers/<Manager_ID>/VirtualMedia/`

2.2.1 Example

Here's an example using cURL.

2.2.1.1 Request

```
$ curl -ku root:0penBmc -X GET 'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/'
```

2.2.1.2 Response


```
{
  "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia",
  "@odata.type": "#VirtualMediaCollection.VirtualMediaCollection",
  "Members": [
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/0"
    },
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/1"
    },
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/2"
    },
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/3"
    },
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/4"
    }
  ],
  "Members@odata.count": 5,
  "Name": "Virtual Media Services"
}
```

2.2.2 Adjust Device Numbers

Users can modify the number of available virtual media devices by sending a **POST** request to: /redfish/v1/Managers/<Manager_ID>/Actions/Oem/InsydeOEMExtensions.VMInstance

Note: The number must be greater than 0 and must not exceed the default value of 5.

2.2.2.1 Example

Here's an example using cURL.

2.2.2.1.1 Request

```
$ curl -ku root:0penBmc -H 'Content-Type: application/json' -X POST
'https://<BMC_IP>/redfish/v1/Managers/bmc/Actions/Oem/InsydeOEMExtensions.VMInstance' -d
'{"VMInstance": 3}'
```

2.2.2.1.2 Response


```
{
 "@Message.ExtendedInfo": [
  {
   "@odata.type": "#Message.v1_1_2.Message",
   "Message": "The request completed successfully.",
   "MessageArgs": [],
   "MessageId": "Base.1.16.Success",
   "MessageSeverity": "OK",
   "Resolution": "None"
  }
 ]
}
```


2.2.2.2 Verify Result

After sending the **POST** request, you can verify the updated number of virtual media devices. As shown in the example below, the number is now set to 3.

2.2.2.2.1 Request

```
$ curl -ku root:0penBmc -X GET 'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/'
```

2.2.2.2.2 Response


```
{
 "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia",
 "@odata.type": "#VirtualMediaCollection.VirtualMediaCollection",
 "Members": [
  {
   "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/0"
  },
  {
   "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/1"
  },
```

```
    {
      "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/2"
    }
  ],
  "Members@odata.count": 3,
  "Name": "Virtual Media Services"
}
```


2.3 Available Operations for Virtual Media

Users can perform three operations on the virtual media devices:

- **Create**: Insert a virtual media.
- **Read**: Read the content of a virtual media.
- **Delete**: Eject a virtual media.

3 Read Virtual Media

Users can read the content of a virtual media device by sending a **GET** request to:
/redfish/v1/Managers/<Manager_ID>/VirtualMedia/<Device_ID>/

3.1 Example

Here's an example using cURL.

3.1.1 Request

```
$ curl -ku root:0penBmc -X GET 'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/0'
```

3.1.2 Response


```
{
  "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/0",
  "@odata.type": "#VirtualMedia.v1_6_4.VirtualMedia",
  "Actions": {
    "#VirtualMedia.EjectMedia": {
      "target": "/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/VirtualMedia.EjectMedia"
    },
    "#VirtualMedia.InsertMedia": {
      "@Redfish.ActionInfo": "/redfish/v1/Managers/bmc/VirtualMedia/0/InsertMediaActionInfo",
      "target": "/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/VirtualMedia.InsertMedia"
    },
    "Oem": {
      "#InsydeOEMExtensions.v1_0_0.PostCertificate": {
        "target":
"/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/Oem/InsydeOEMExtensions.PostCertificate"
      }
    }
  },
  "Certificates": {
    "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/Oem/Insyde/Certificates/0"
  },
  "ConnectedVia": "URI",
  "Id": "0",
  "Image": "https://releases.ubuntu.com/noble/ubuntu-24.04.2-desktop-amd64.iso",
  "Inserted": true,
  "MediaTypes": [
    "CD",
    "DVD"
  ],
  "Name": "Virtual Removable Media",
  "Password": null,
  "Status": {
    "Health": "OK",
    "HealthRollup": "OK",
    "State": "Enabled"
  },
```

```
  "TransferMethod": "Stream",
  "TransferProtocolType": "HTTPS",
  "UserName": "",
  "VerifyCertificate": true,
  "WriteProtected": true
}
```


3.2 Essential Properties

Property	Description
Certificates	The link to a collection of server certificates for the server referenced by the **Image** property.
Image	The URI of the location of the selected image.
Inserted	Indicates whether virtual media is inserted into the virtual device.
TransferProtocolType	The network protocol to use with the URI specified by the **Image** property.
VerifyCertificate	Indicates whether the service will verify the certificate of the server referenced by the **Image** property prior to completing the remote media connection.

Note: For detailed descriptions of additional properties, please refer to *Supervyse OPF Redfish API User Guide*.

Note: For detailed information on the usage of **Certificates**, please refer to *Supervyse OPF Redfish CertificateService User Guide*.

4 Insert Virtual Media

4.1 Insert Media Action Info

Users can read the parameter requirements for the InsertMedia action by sending a **GET** request to: /redfish/v1/Managers/<Manager_ID>/VirtualMedia/<Device_ID>/InsertMediaActionInfo

4.1.1 Example

Here's an example using cURL.

4.1.1.1 Request

```
$ curl -ku root:0penBmc -X GET
'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/0/InsertMediaActionInfo'
```

4.1.1.2 Response


```
{
 "@odata.id": "/redfish/v1/Managers/bmc/VirtualMedia/0/InsertMediaActionInfo",
 "@odata.type": "#ActionInfo.v1_2_0.ActionInfo",
 "Id": "InsertMediaActionInfo",
 "Name": "Virtual Media Insert Media Action Info",
 "Parameters": [
  {
   "DataType": "String",
   "Name": "Image",
   "Required": true
  },
  {
   "DataType": "String",
   "Name": "UserName",
   "Required": false
  },
  {
   "DataType": "String",
   "Name": "Password",
   "Required": false
  },
  {
   "DataType": "String",
   "Name": "TransferProtocolType",
   "Required": false,
   "TransferProtocolType@Redfish.AllowableValues": [
    "CIFS",
    "NFS",
    "HTTPS",
    "OEM"
   ]
  }
 ]
}
```

4.2 Insert Media

Users can attach remote media to the system by sending a **POST** request to:
/redfish/v1/Managers/<Manager_ID>/VirtualMedia/<Device_ID>/Actions/VirtualMedia.InsertMedia

4.2.1 Action Parameters

Property	Type	Description	Attributes
Image	string	The URI of the media to attach to the virtual media.	required
UserName	string	The username to access the URI specified by the **Image** parameter.	optional
Password	string	The password to access the URI specified by the **Image** parameter.	optional
TransferProtocolType	string	The network protocol to use with the URI specified by the **Image** parameter.	optional
VerifyCertificate	boolean	An indicator of whether the service will verify the certificate of the server referenced by the **Image** parameter prior to completing the remote media connection.	optional

4.2.1.1 Applicable TransferProtocolType

string	Description
CIFS	Common Internet File System.
NFS	Network File System.
HTTPS	Hypertext Transfer Protocol Secure.
OEM	A manufacturer-defined protocol.

4.2.1.2 Specifying the Transfer Protocol

The Virtual Media service can automatically parse the appropriate transfer protocol from the Image URI. However, it is **recommended to explicitly specify the protocol** using the TransferProtocolType parameter.

4.2.2 Example (HTTPS)

Here's an example via an HTTPS server using cURL.

4.2.2.1 Request

```
$ curl -ku root:0penBmc -H "Content-Type: application/json" -X POST
'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/VirtualMedia.InsertMedia' -d
```

```
'{"Image": "https://192.168.1.225/boot_image.iso", "TransferProtocolType": "HTTPS",
"VerifyCertificate": false}'
```

4.2.2.2 Response


```
{
  "@Message.ExtendedInfo": [
    {
      "@odata.type": "#Message.v1_1_2.Message",
      "Message": "The request completed successfully.",
      "MessageArgs": [],
      "MessageId": "Base.1.16.Success",
      "MessageSeverity": "OK",
      "Resolution": "None"
    }
  ]
}
```


4.2.3 Example (CIFS)

Here's an example via an CIFS server using cURL.

4.2.3.1 Request

```
$ curl -ku root:0penBmc -H "Content-Type: application/json" -X POST
'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/VirtualMedia.InsertMedia' -d
'{"Image": "cifs//192.168.1.225/smbfolder/boot_image.iso", "UserName": "smbuser", "Password":
"smbpassword", "TransferProtocolType": "CIFS", "VerifyCertificate": false}'
```

4.2.3.2 Response


```
{
  "@Message.ExtendedInfo": [
    {
      "@odata.type": "#Message.v1_1_2.Message",
      "Message": "The request completed successfully.",
      "MessageArgs": [],
      "MessageId": "Base.1.16.Success",
      "MessageSeverity": "OK",
      "Resolution": "None"
    }
  ]
}
```

5 Eject Virtual Media

Users can detach remote media from the system by sending a **POST** request to:
/redfish/v1/Managers/<Manager_ID>/VirtualMedia/<Device_ID>/Actions/VirtualMedia.EjectMedia

5.1 Example

Here's an example using cURL.

5.1.1 Request

```
$ curl -ku root:0penBmc -X POST
'https://<BMC_IP>/redfish/v1/Managers/bmc/VirtualMedia/0/Actions/VirtualMedia.EjectMedia'
```

5.1.2 Response


```
{
 "@Message.ExtendedInfo": [
  {
   "@odata.type": "#Message.v1_1_2.Message",
   "Message": "The request completed successfully.",
   "MessageArgs": [],
   "MessageId": "Base.1.16.Success",
   "MessageSeverity": "OK",
   "Resolution": "None"
  }
 ]
}
```